

TOSHIBA CMOS DIGITAL INTEGRATED CIRCUIT SILICON MONOLITHIC

TC74VHCT240F, TC74VHCT240FW, TC74VHCT240FS
TC74VHCT244F, TC74VHCT244FW, TC74VHCT244FS
OCTAL BUS BUFFER**TC74VHCT240F/FW/FS INVERTED, 3-STATE OUTPUTS****TC74VHCT244F/FW/FS NON-INVERTED, 3-STATE OUTPUTS**

The TC74VHCT240 and 244 are advanced high speed CMOS OCTAL BUS BUFFERS fabricated with silicon gate C²MOS technology. They achieve the high speed operation similar to equivalent Bipolar Schottky TTL while maintaining the CMOS low power dissipation.

The TC74VHCT240 is an inverting 3-state buffer having two active - low output enables. The TC74VHCT244 is a non-inverting 3-state buffer, and has two active - low output enables.

These devices are designed to be used with 3-state memory address drivers, etc.

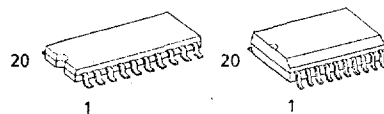
The input voltage are compatible with TTL output voltage.

These devices may be used as a level converter for interfacing 3.3V to 5V system.

Input protection and output circuit ensure that 0 to 7V can be applied to the input and output pins without regard to the supply voltage. These structure prevents device destruction due to mismatched supply and input/output voltages such as battery back up, hot board insertion, etc.

FEATURES:

- High Speed..... $t_{pd} = 5.6\text{ns}(\text{typ.})$ at $V_{CC} = 5\text{V}$
- Low Power Dissipation..... $I_{CC} = 4\mu\text{A}(\text{Max.})$ at $T_a = 25^\circ\text{C}$
- Compatible with TTL outputs ... $V_{IL} = 0.8\text{V}(\text{Max.})$
 $V_{IH} = 2.0\text{V}(\text{Min.})$
- Power Down Protection is provided on all inputs and outputs
- Balanced Propagation Delays..... $t_{PLH} \approx t_{PHL}$
- Low Noise $V_{OLP} = 1.1\text{V}(\text{Max.})$
- Pin and Function Compatible with 74ALS240/244

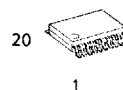


F (SOP20-P-300-1.27)

Weight : 0.22g (TYP.)

FW (SOL20-P-300-1.27)

Weight : 0.46g (TYP.)



FS (SSOP20-P-225-0.65A)

Weight : 0.09g (TYP.)

APPLICATION NOTE

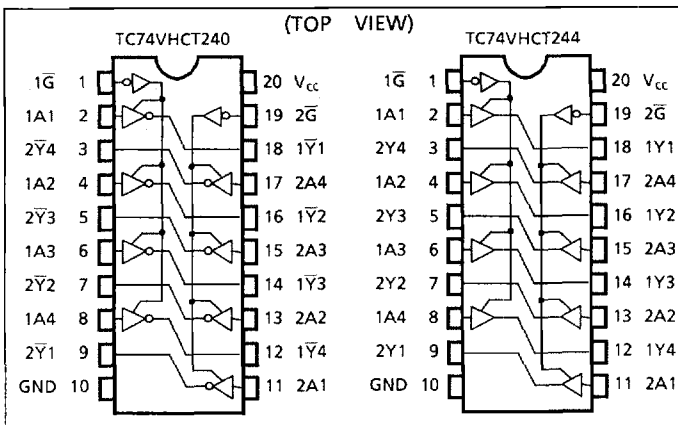
These devices can drive components with CMOS input level by adding a external pull up resistor to output terminal.

TRUTH TABLE

INPUTS		OUTPUTS	
$\overline{G}$	A_n	Y_n	$\overline{Y}_n$
L	L	L	H
L	H	H	L
H	X	Z	Z

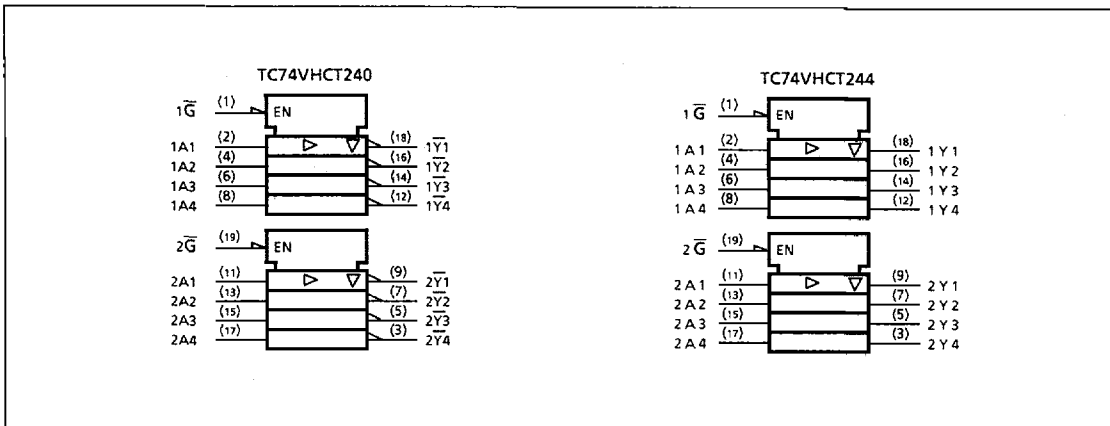
X : Don't Care

Z : High Impedance

 Y_n : TC74VHCT244 $\overline{Y}_n$: TC74VHCT240**PIN ASSIGNMENT**

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IEC LOGIC SYMBOL



ABSOLUTE MAXIMUM RATINGS

PARAMETER	SYMBOL	VALUE	UNIT
Supply Voltage Range	V_{CC}	-0.5~7.0	V
DC Input Voltage	V_{IN}	-0.5~7.0	V
DC Output Voltage	V_{OUT}	-0.5~7.0	V
Input Diode Current	I_{IK}	-20	mA
Output Diode Current	I_{OK}	-20	mA
DC Output Current	I_{OUT}	± 25	mA
DC V_{CC} /Ground Current	I_{CC}	± 75	mA
Power Dissipation	P_D	180	mW
Storage Temperature	T_{stg}	-65~150	°C

RECOMMENDED OPERATING CONDITIONS

PARAMETER	SYMBOL	VALUE	UNIT
Supply Voltage	V_{CC}	4.5~5.5	V
Input Voltage	V_{IN}	0~5.5	V
Output Voltage	V_{OUT}	0~5.5	V
Operating Temperature	T_{opr}	-40~85	°C
Input Rise and Fall Time	dt/dV	0~20	ns/V

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DC ELECTRICAL CHARACTERISTICS

PARAMETER	SYMBOL	CONDITON	$V_{CC}(V)$	$T_a = 25^\circ C$			$T_a = -40 \sim 85^\circ C$		UNIT
				MIN.	TYP.	MAX.	MIN.	MAX.	
High - Level Input Voltage	V_{IH}		4.5~5.5	2.0	—	—	2.0	—	V
Low - Level Input Voltage	V_{IL}		4.5~5.5	—	—	0.8	—	0.8	V
High - Level Output Voltage	V_{OH}	$V_{IN} = V_{IH} \text{ or } V_{IL}$	$I_{OH} = -50 \mu A$	4.5	3.15	3.65	—	3.15	V
			$I_{OH} = -8 mA$	4.5	2.50	—	—	2.40	
Low - Level Output Voltage	V_{OL}	$V_{IN} = V_{IH} \text{ or } V_{IL}$	$I_{OL} = 50 \mu A$	4.5	—	0.0	—	0.10	V
			$I_{OL} = 8 mA$	4.5	—	—	—	0.44	
3 - State Output Off - State Current	I_{OZ}	$V_{IN} = V_{IH} \text{ or } V_{IL}$ $V_{OUT} = V_{CC} \text{ or } GND$	5.5	—	—	± 0.25	—	± 2.50	μA
Input Leakage Current	I_{IN}	$V_{IN} = 5.5V \text{ or } GND$	0~5.5	—	—	± 0.1	—	± 1.0	
Quiescent Supply Current	I_{CC}	$V_{IN} = V_{CC} \text{ or } GND$	5.5	—	—	4.0	—	40.0	
	I_{CCT}	PER INPUT : $V_{IN} = 3.4V$ OTHER INPUT : $V_{CC} \text{ or } GND$	5.5	—	—	1.35	—	1.50	mA
Output Leakage Current	I_{OPD}	$V_{OUT} = 5.5V$	0	—	—	+0.5	—	+5.0	μA

AC ELECTRICAL CHARACTERISTICS (Input $t_r = t_f = 3ns$)

PARAMETER	SYMBOL	TEST CONDITION		$T_a = 25^\circ C$			$T_a = -40 \sim 85^\circ C$		UNIT
		$V_{CC}(V)$	$C_L(pF)$	MIN.	TYP.	MAX.	MIN.	MAX.	
Propagation Delay Time (TC74VHCT240)	t_{PLH}	5.0 $\pm$ 0.5	15	—	5.6	7.8	1.0	9.0	ns
	t_{PHL}		50	—	6.1	8.8	1.0	10.0	
Propagation Delay Time (TC74VHCT244)	t_{PLH}	5.0 $\pm$ 0.5	15	—	5.4	7.4	1.0	8.5	
	t_{PHL}		50	—	5.9	8.4	1.0	9.5	
3-State Output Enable Time	t_{PZL}	$R_L = 1k\Omega$	5.0 $\pm$ 0.5	15	—	7.7	10.4	1.0	
	t_{PZH}		50	—	8.2	11.4	1.0	13.0	
3-State Output Disable Time	t_{PLZ} t_{PHZ}	$R_L = 1k\Omega$	5.0 $\pm$ 0.5	50	—	8.8	11.4	1.0	13.0
Output to Output Skew	t_{OSLH} t_{OSHL}	(Note 1)	5.0 $\pm$ 0.5	50	—	—	1.0	—	1.0
Input Capacitance	C_{IN}			—	4	10	—	10	pF
Output Capacitance	C_{OUT}			—	9	—	—	—	
Power Dissipation Capacitance (Note 2)	C_{PD}	TC74VHCT240		—	19	—	—	—	
		TC74VHCT244		—	18	—	—	—	

Note (1) Parameter guaranteed by design. $t_{OSLH} = t_{pLHm} - t_{pLHn}$, $t_{OSHL} = t_{pHLm} - t_{pHLn}$

Note (2) C_{PD} is defined as the value of the internal equivalent capacitance which is calculated from the operating current consumption without load.

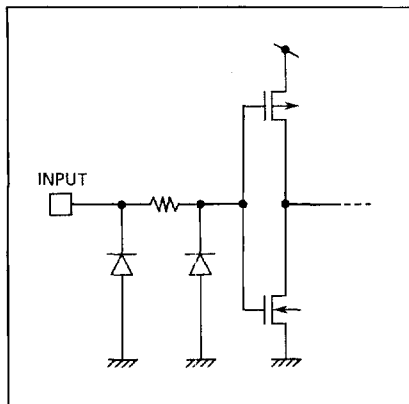
Average operating current can be obtained by the equation :

$$I_{CC(opr.)} = C_{PD} \cdot V_{CC} \cdot f_{IN} + I_{CC}/8 \text{ (per bit)}$$

NOISE CHARACTERISTICS (Input $t_r = t_f = 3\text{ns}$)

PARAMETER	SYMBOL	TEST CONDITION	Ta = 25°C		UNIT
			V _{CC} (V)	TYP.	LIMIT
Quiet Output Maximum Dynamic V _{OL}	V _{OLP}	C _L = 50pF	5.0	0.9	1.1
Quiet Output Minimum Dynamic V _{OL}	V _{OLV}	C _L = 50pF	5.0	-0.9	-1.1
Minimum High Level Dynamic Input Voltage	V _{IHD}	C _L = 50pF	5.0	—	2.0
Maximum Low Level Dynamic Input Voltage	V _{ILD}	C _L = 50pF	5.0	—	0.8

INPUT EQUIVALENT CIRCUIT



OUTPUT EQUIVALENT CIRCUIT

